01 Remote Connectivity and File Transfer

1. Preparation

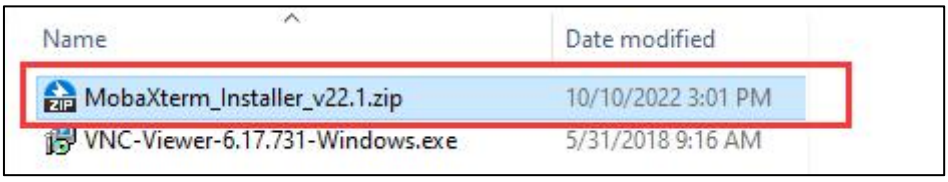
1.1 Hardware Preparation

Prepare a laptop for the setup. If you're using a desktop computer, make sure to have a wireless network adapter, as it is necessary to connect both the desktop and the Raspberry Pi 5 to the same local network.

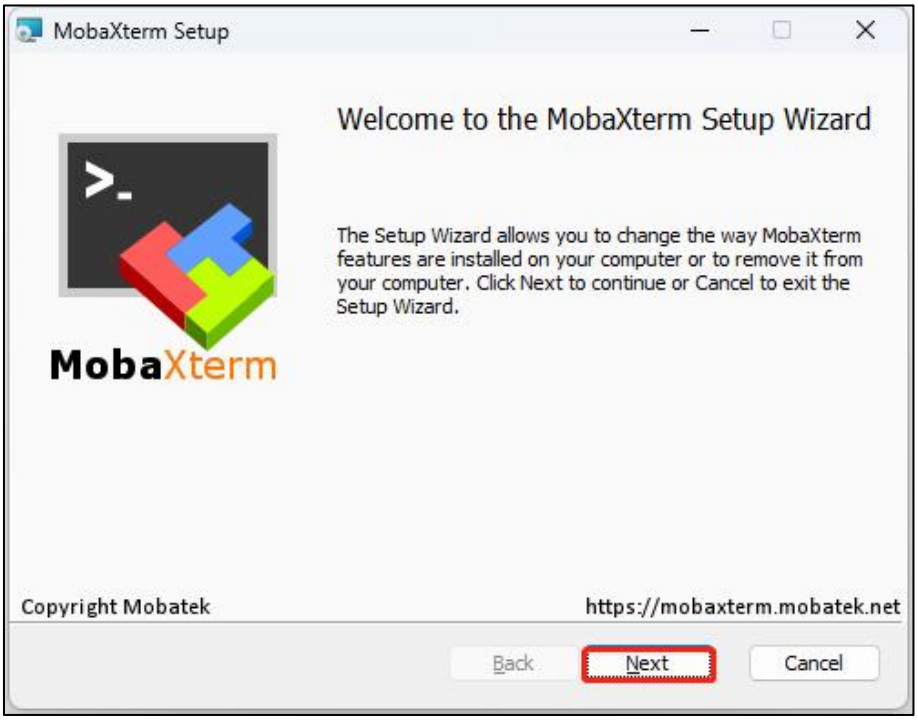
1.2 MobaXterm Tool Installation and Usage

MobaXterm is a software that integrates various remote computing tools and provides a graphical interface. By connecting to the Raspberry Pi's Wi-Fi hotspot, you can control the Raspberry Pi directly from your computer. Follow the steps below to install the software:

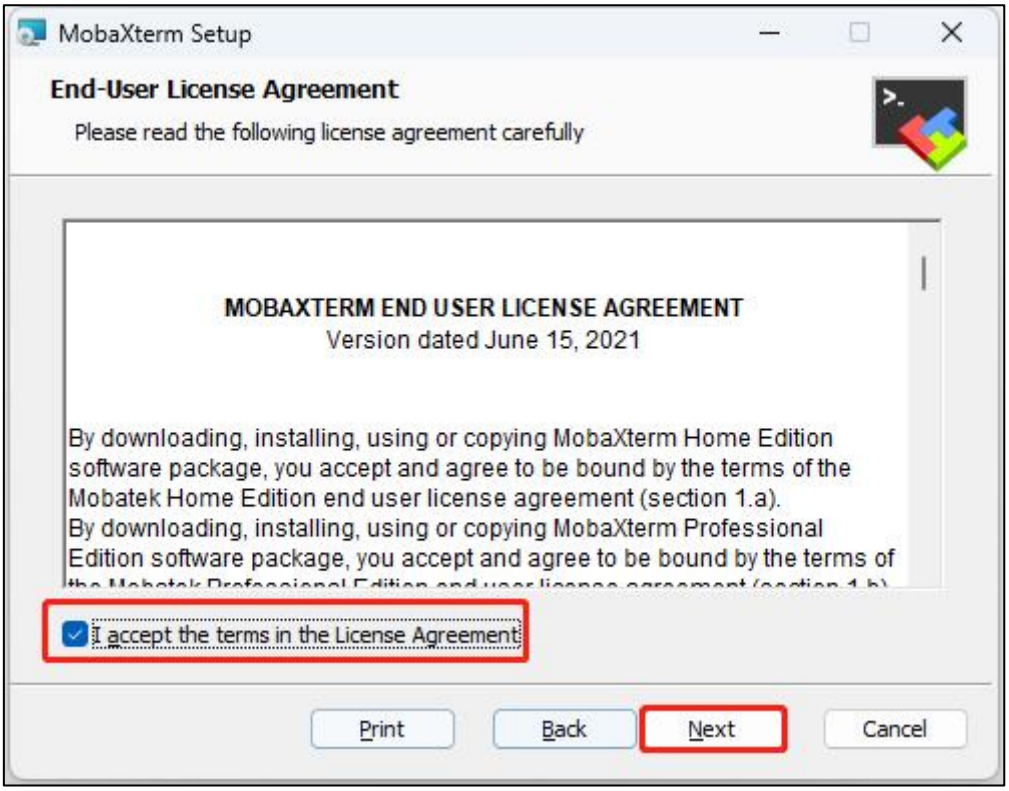
1) Extract the "**MobaXterm_Installer_v22.1.zip**" file located in the "4.6 Raspberry Pi Communication\01 Remote Connectivity and File Transfer" folder.



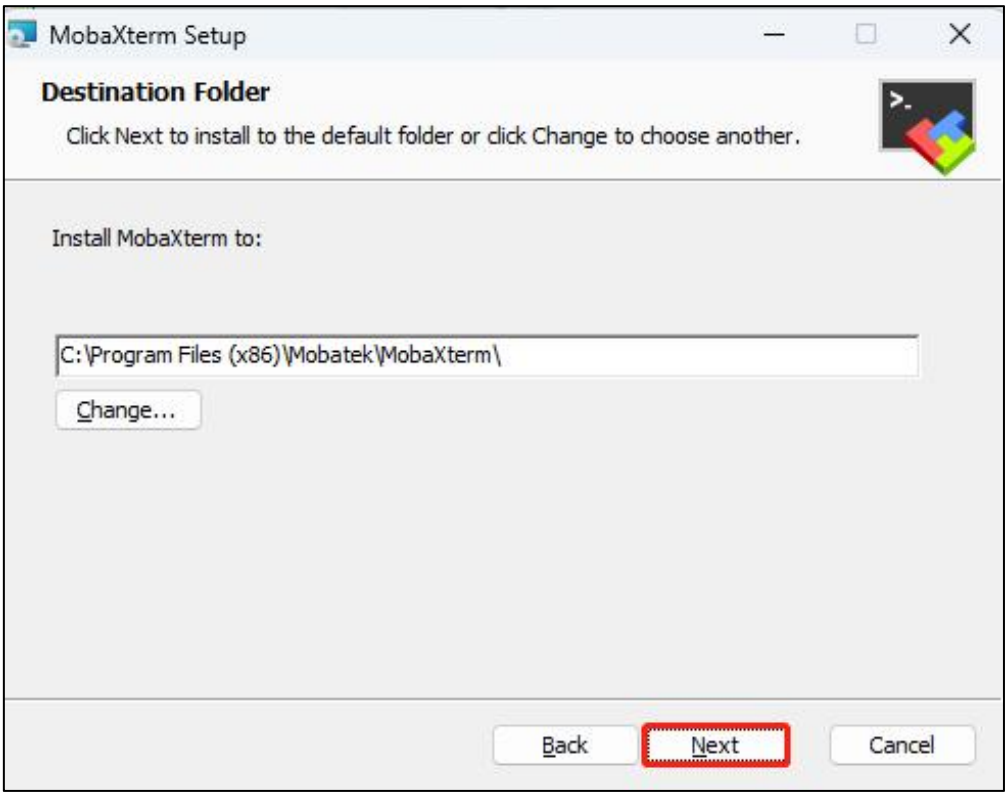
2) Double-click the "MobaXterm_installer_22.1.msi" file in the extracted "MobaXterm_Installer_v22.1" folder and click "Next."



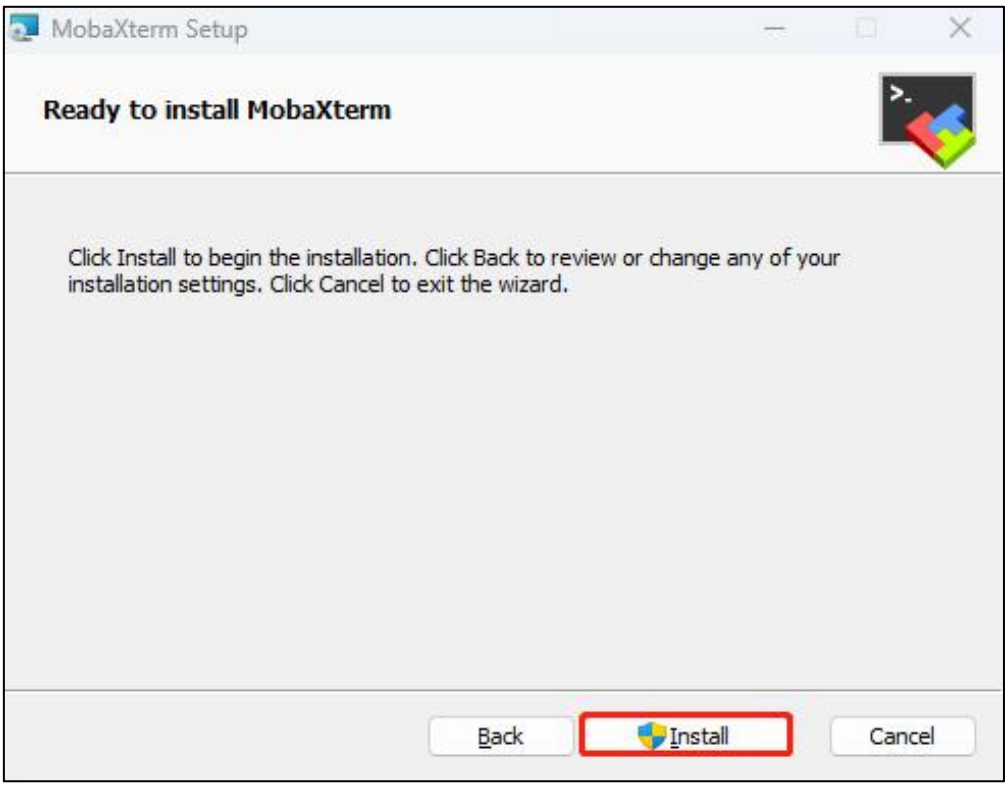
3) In the pop-up window, accept the license agreement and click "Next."



4) Choose the installation path and click "Next."

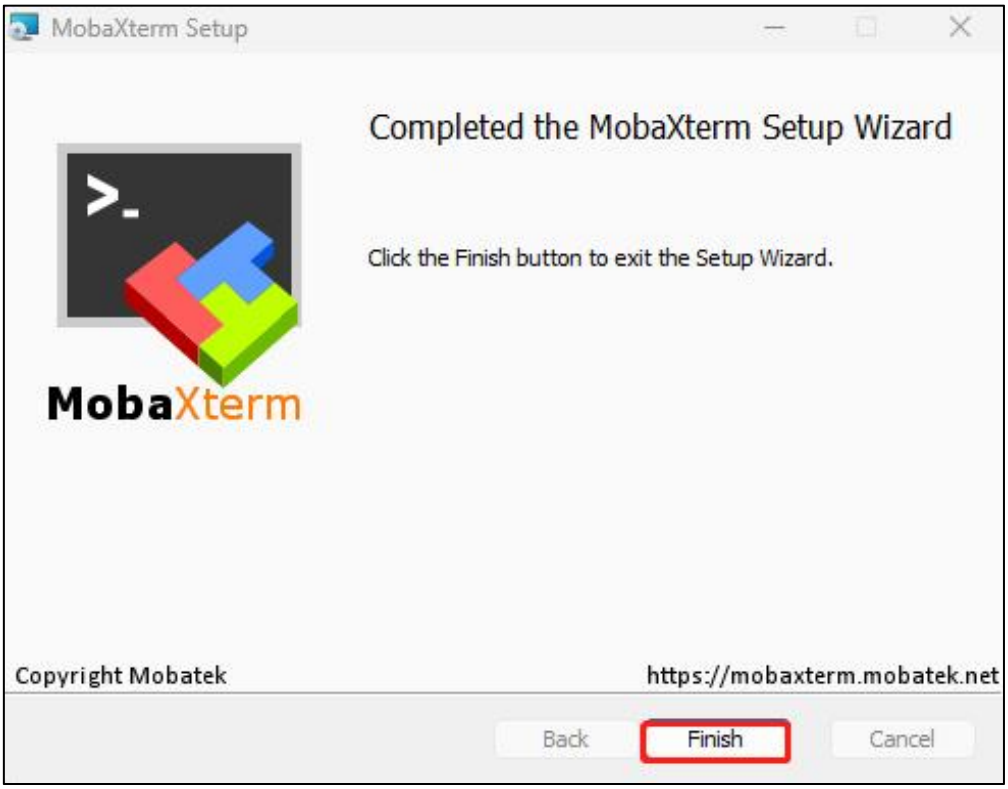


5) Click "Install."



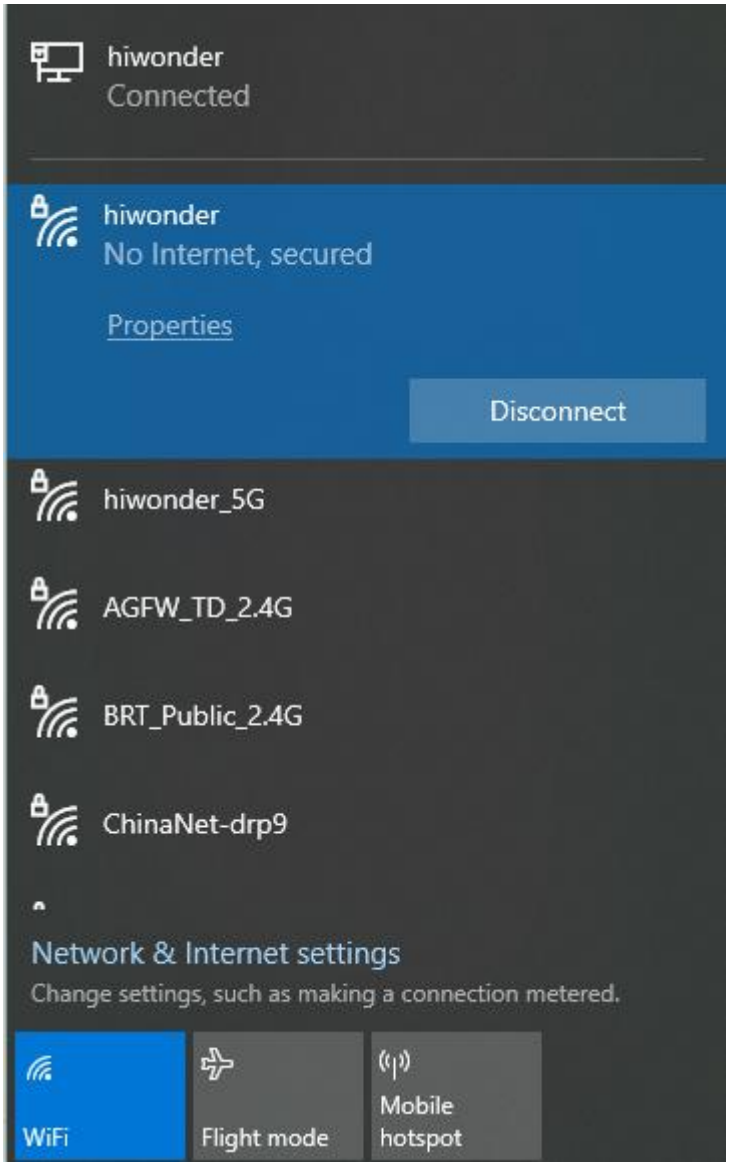
6) Wait for the installation to complete. Once finished, a prompt will appear

indicating the installation is complete. Click the "Finish" button to complete the process.

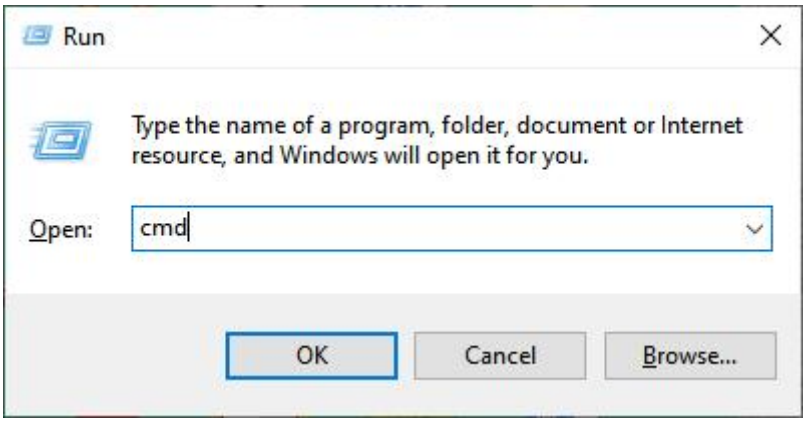


1.3 Start Raspberry Pi 5

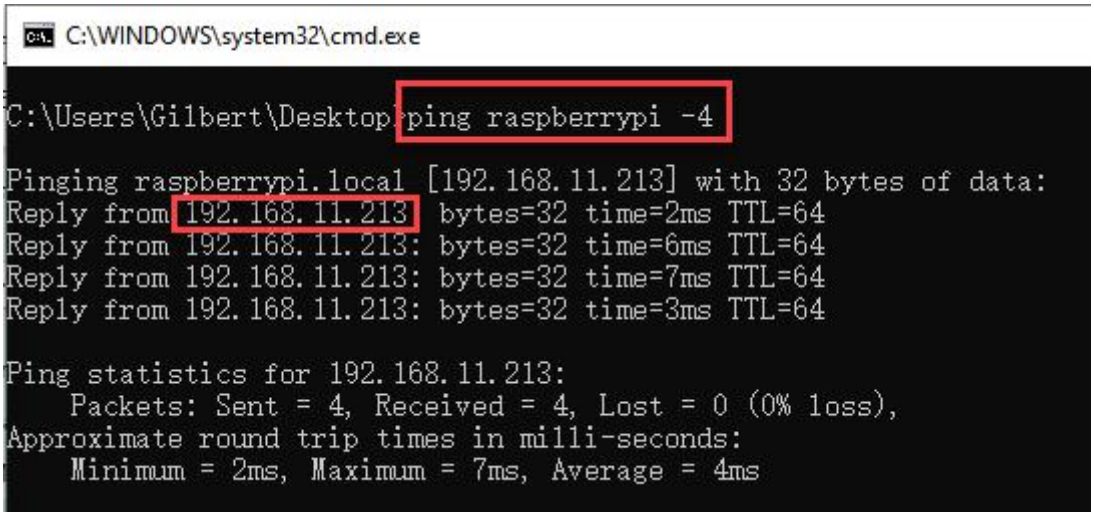
1) Connect the Raspberry Pi 5 to the local network. In this tutorial, the Raspberry Pi 5 is already connected to the "**Hiwonder**" hotspot. Next, connect your computer to the same "**Hiwonder**" hotspot.



2) Press Win + R to open the "Run" dialog, type "cmd" and press Enter to open the Command Prompt in administrator mode.

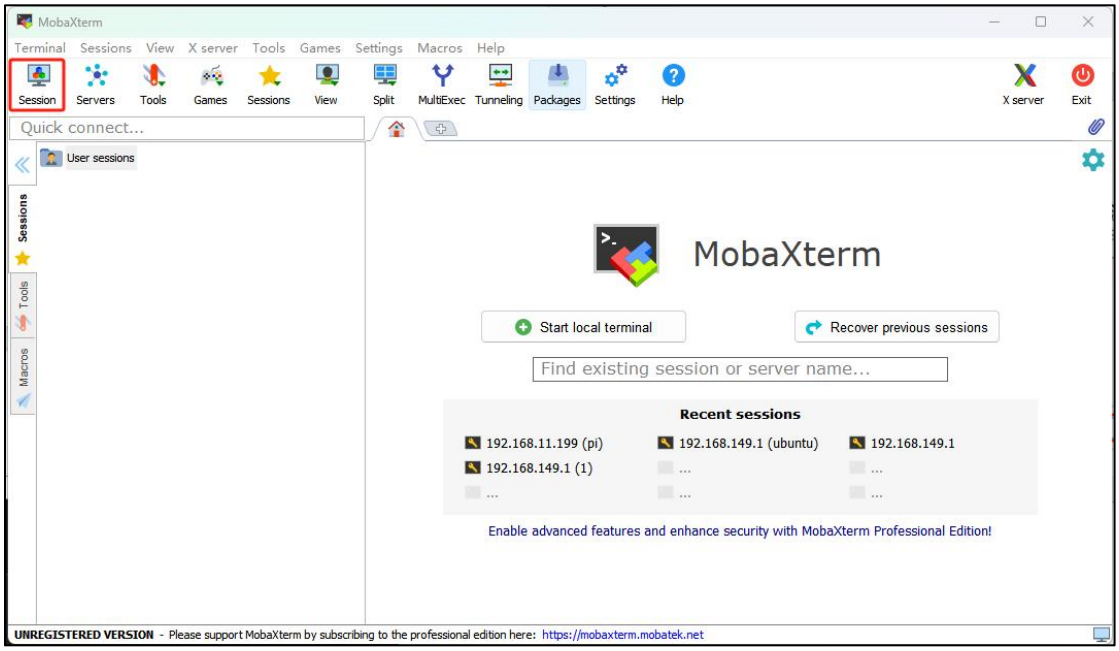


3) Type the command "**ping raspberrypi -4**" and press Enter to find the IP address assigned to the Raspberry Pi 5 by the hotspot.



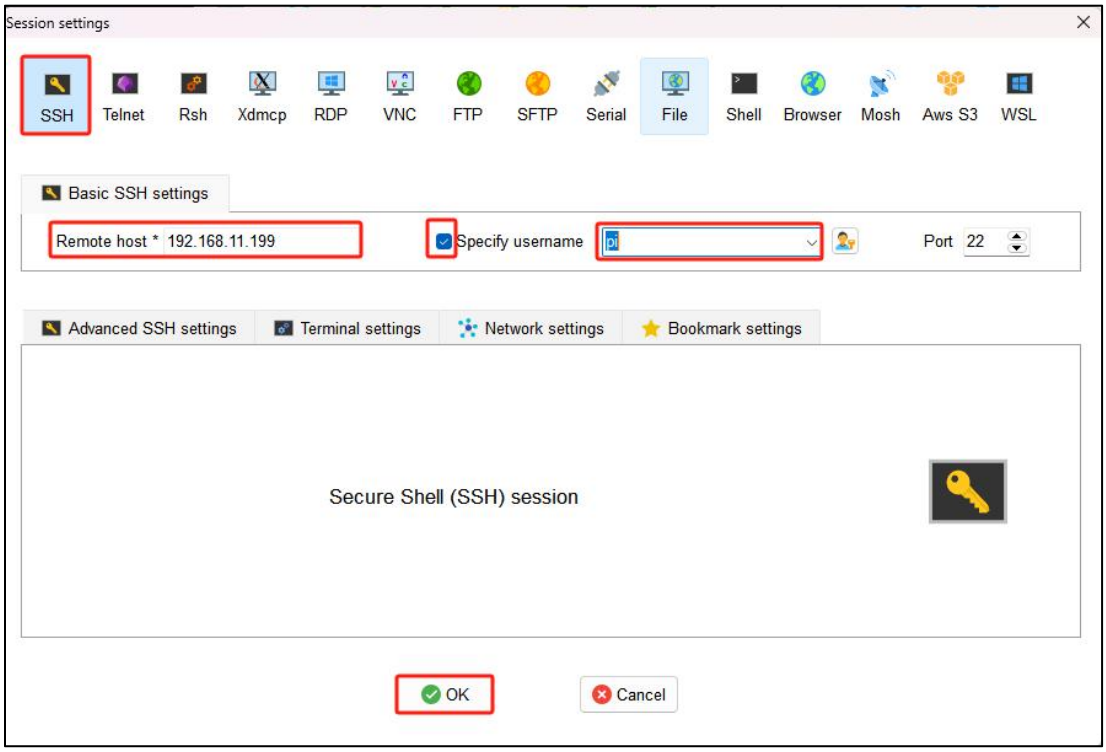
2. Connect to Raspberry Pi 5 using MobaXterm

1) Open MobaXterm. In the main interface, click on the "**Session**" button in the top-left corner to create a new session.

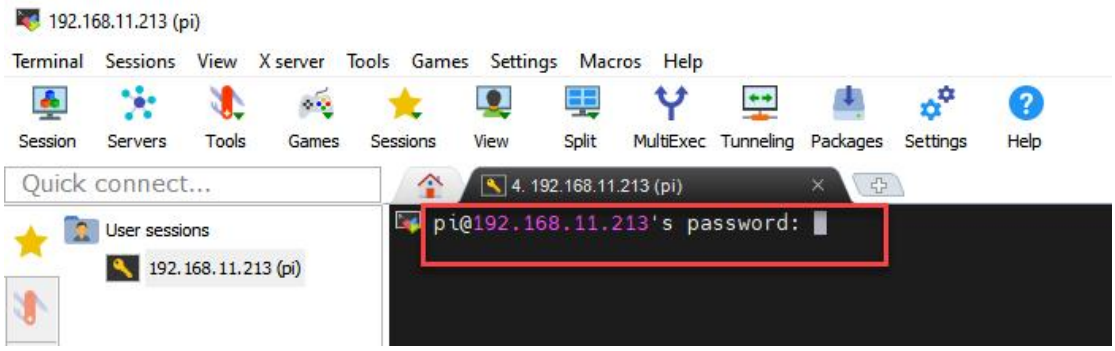


2) In the session window, select "SSH." Enter the Raspberry Pi's IP address ("**192.168.11.199**"), check the "Specify username" box, and input the

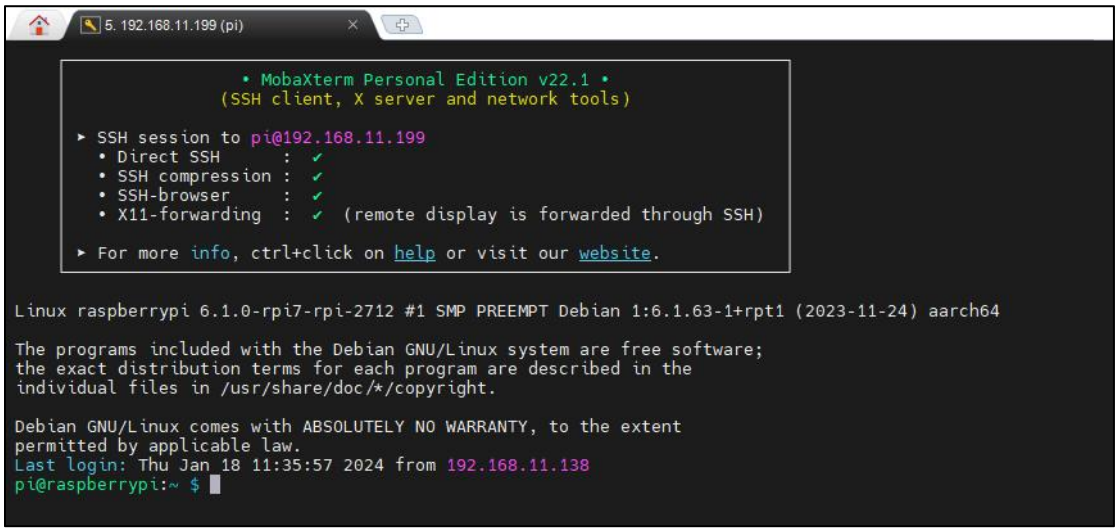
username "pi." Then, click "OK."



3) On the first connection, you will be prompted to enter the Raspberry Pi 5's password, which is "raspberrypi." The password will not be visible as you type. After entering it correctly, press Enter.



4) Once the password is verified, you will successfully log into the system. The system interface will appear as shown in the image below:



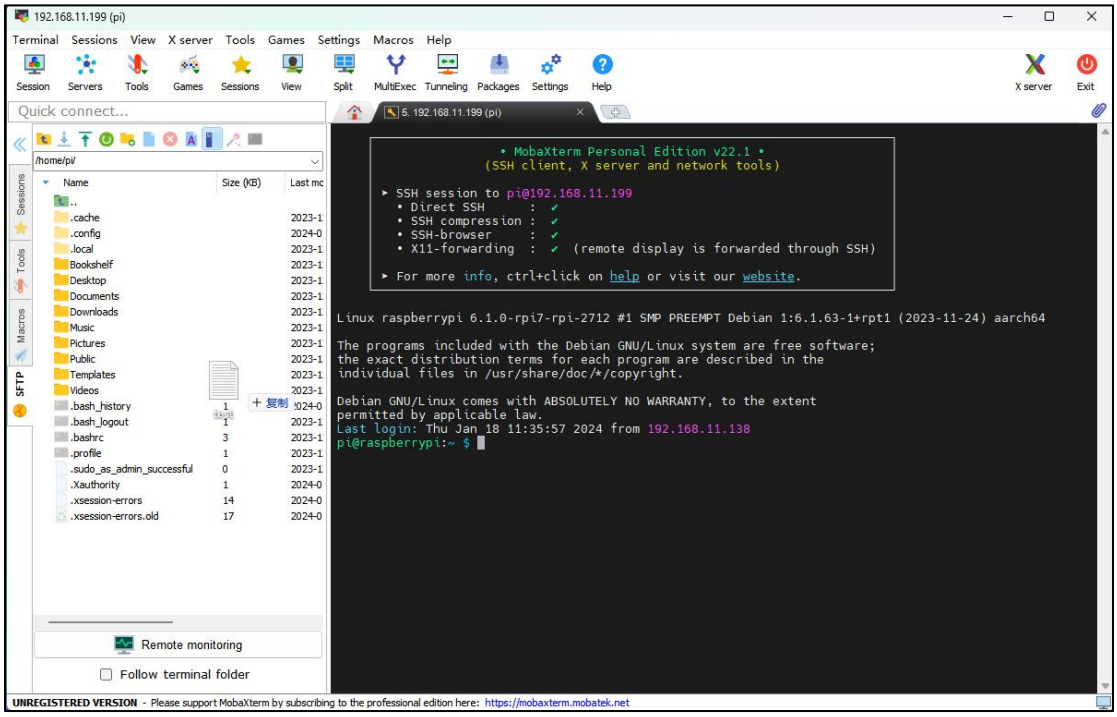
3. File Transfer Between Raspberry Pi and Computer

Transferring files between your computer and Raspberry Pi is a common task. In this section, we'll demonstrate how to do this using SSH and MobaXterm. For other file transfer methods, you can search online for further guidance.

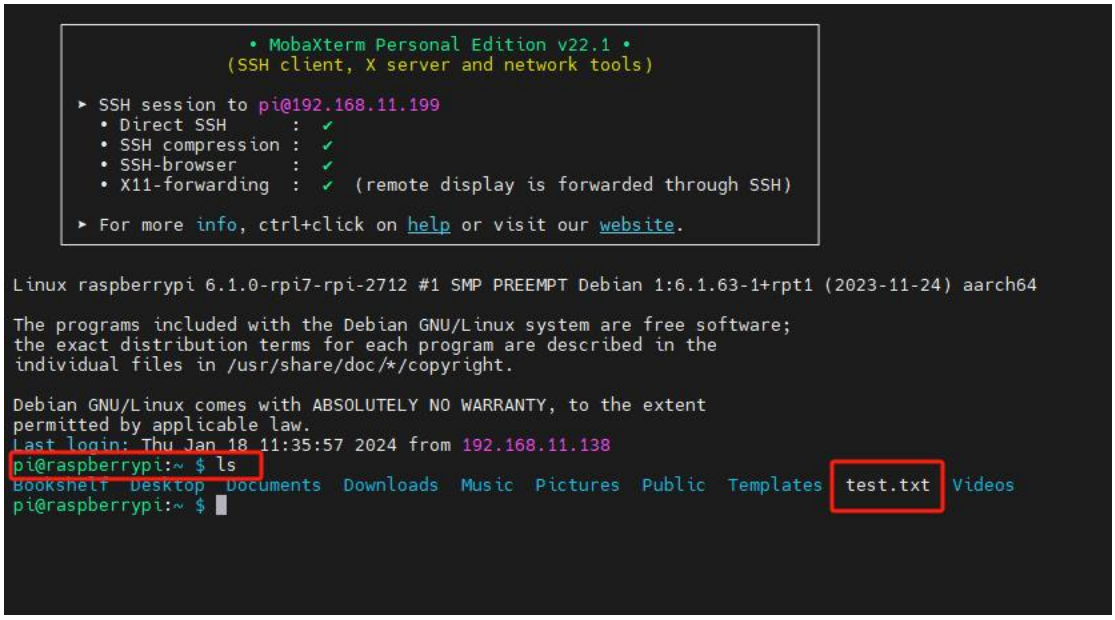
3.1 Transfer Files from Computer to Raspberry Pi

1) After establishing the SSH connection, create a "test.txt" file on your computer’s desktop as an example.

2) Adjust the zoom level in MobaXterm and select the file you wish to transfer. Drag and drop the file into the MobaXterm file area.



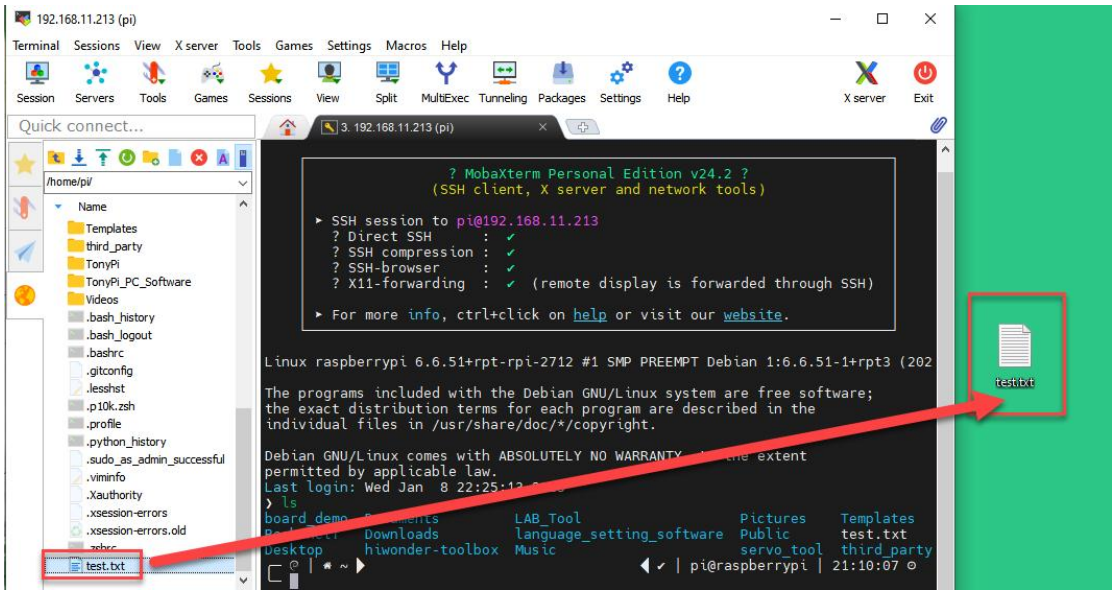
3) The file will automatically be transferred to the Raspberry Pi desktop. To view all files, type the "ls" command and press Enter. As shown in the image below, the file has been successfully transferred from the computer.



3.2 Transfer Files from Raspberry Pi to Computer

After establishing the SSH connection, navigate to the file you want to transfer

in the MobaXterm file area. Drag and drop the file onto your computer's desktop, and it will be automatically transferred.



4. Enable VNC Connection

If you're not comfortable with the command line or prefer an alternative, the official Raspberry Pi 5 system image doesn't enable VNC by default. You can enable it through an SSH connection.

1) After connecting via SSH, type **sudo raspi-config** and press Enter to open the Raspberry Pi configuration menu.

```
                • MobaXterm Personal Edition v22.1 •
             (SSH client, X server and network tools)

 ➤ SSH session to pi@192.168.11.199
   • Direct SSH      :  ✔
   • SSH compression :  ✔
   • SSH-browser     :  ✔
   • X11-forwarding  :  ✔  (remote display is forwarded through SSH)

 ➤ For more info, ctrl+click on help or visit our website.

Linux raspberrypi 6.1.0-rpi7-rpi-2712 #1 SMP PREEMPT Debian 1:6.1.63-1+rpt1 (2023-11-24) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jan 18 11:38:54 2024 from 192.168.11.138
pi@raspberrypi:~ $ sudo raspi-config
```

2) Use the "↑" and "↓" arrow keys to navigate, "Enter" to confirm a selection, and "Esc" to go back. Select "Interface Options" and press Enter to proceed to the related settings.

```
Raspberry Pi Software Configuration Tool (raspi-config)

 1 System Options        Configure system settings
 2 Display Options       Configure display settings
 3 Interface Options     Configure connections to peripherals
 4 Performance Options   Configure performance settings
 5 Localisation Options  Configure language and regional settings
 6 Advanced Options      Configure advanced settings
 8 Update                Update this tool to the latest version
 9 About raspi-config    Information about this configuration tool

        <Select>                          <Finish>
```

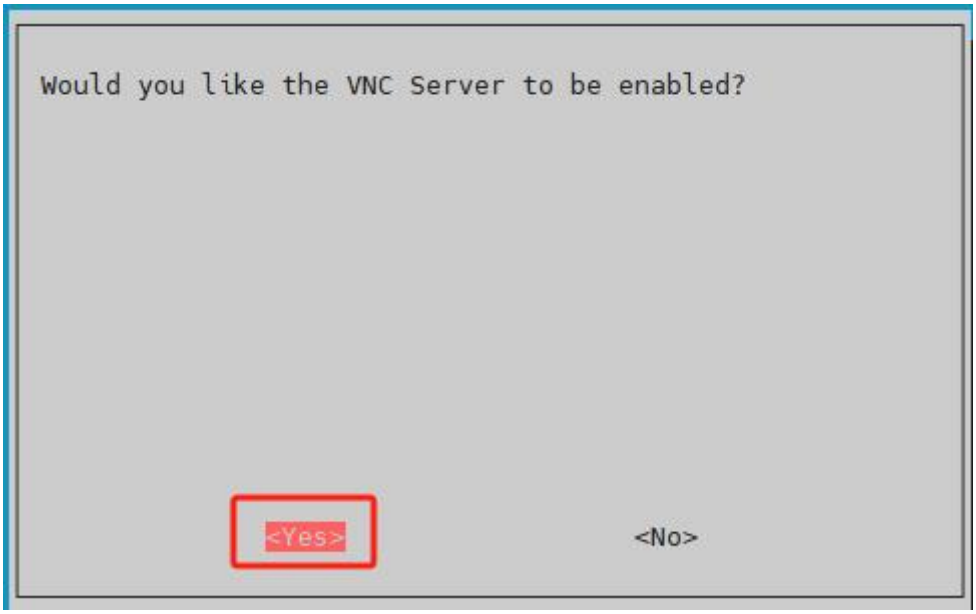
3) Locate "I2 VNC" and press Enter to access the VNC switch screen.

```
Raspberry Pi Software Configuration Tool (raspi-config)

 I1 SSH         Enable/disable remote command line access using SSH
 I2 VNC         Enable/disable graphical remote desktop access
 I3 SPI         Enable/disable automatic loading of SPI kernel module
 I4 I2C         Enable/disable automatic loading of I2C kernel module
 I5 Serial Port Enable/disable shell messages on the serial connection
 I6 1-Wire      Enable/disable one-wire interface
 I7 Remote GPIO Enable/disable remote access to GPIO pins

        <Select>                          <Back>
```

4) Select "Yes" and press Enter to enable the VNC connection.



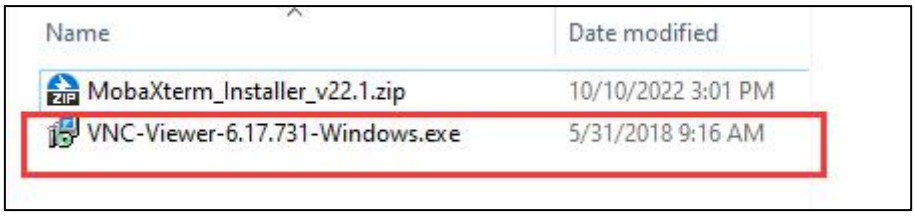
After enabling VNC, you can connect to the Raspberry Pi 5 system via VNC.

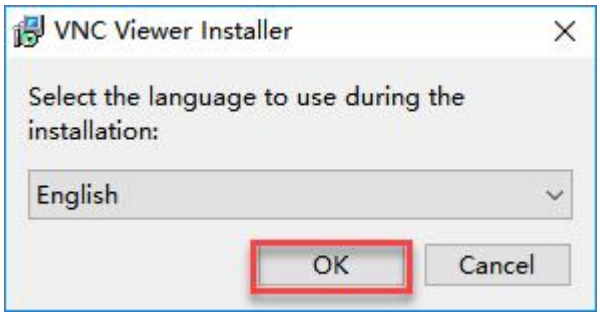
5. VNC Installation and Connection

VNC is a remote desktop software that enables you to control your system remotely through a graphical interface, similar to the Windows desktop.

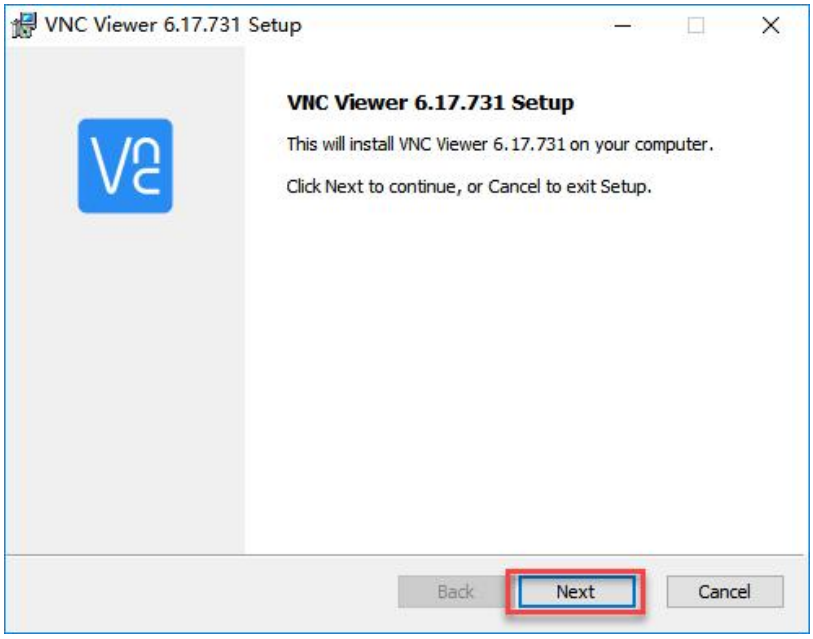
5.1 Installing VNC

1) Double-click the "**VNC-Viewer-6.17.731-Windows**" file in the "4.6.1 Remote Connectivity and File Transfer\02 Remote Connectivity and File Transfer" folder. In the pop-up dialog, select "English" as the installation language and click "OK."

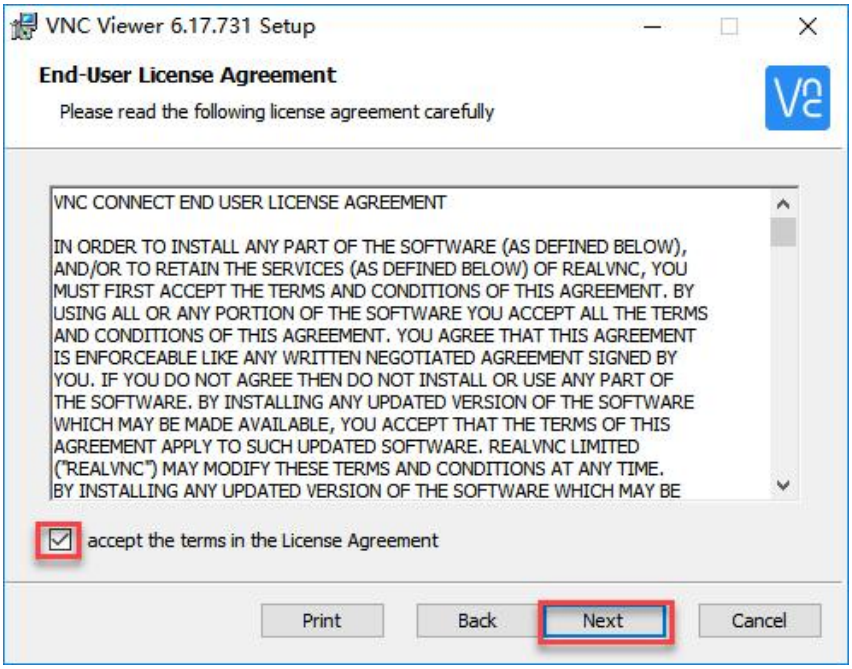




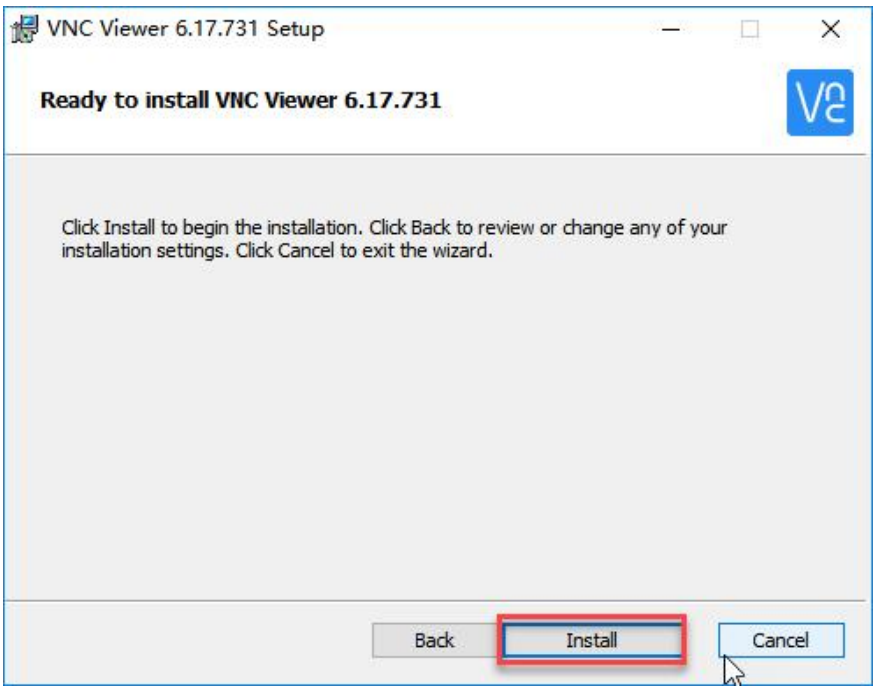
2) Click the “Next” button in the following window.



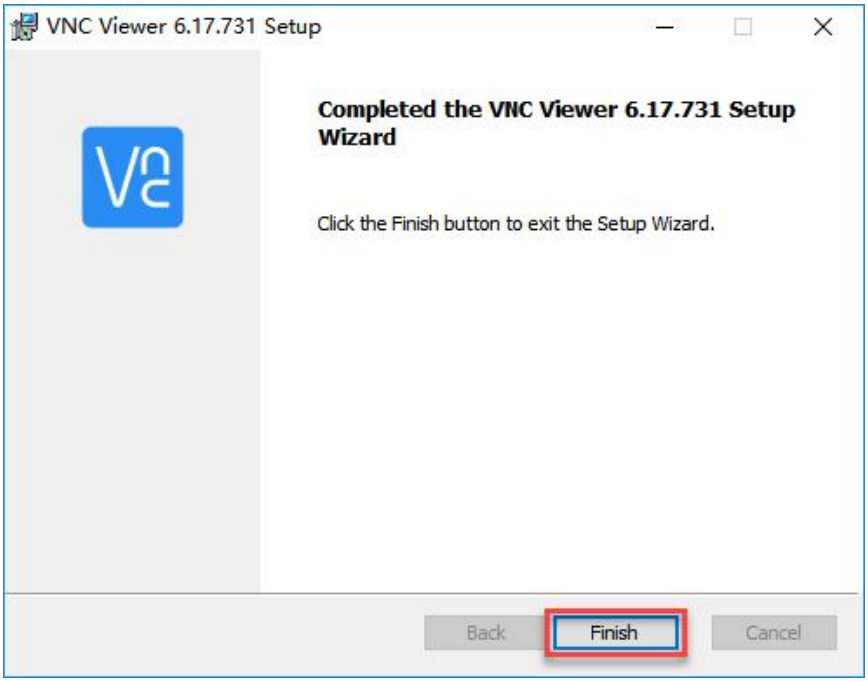
3) In the next prompt, agree to the license agreement and click “Next.” When the installation location settings appear, keep the default and click “Next” again.



4) Click the “Install” button in the next window.



5) Wait for the installation to finish. Once complete, a confirmation message will appear. Click “Finish” to complete the installation.

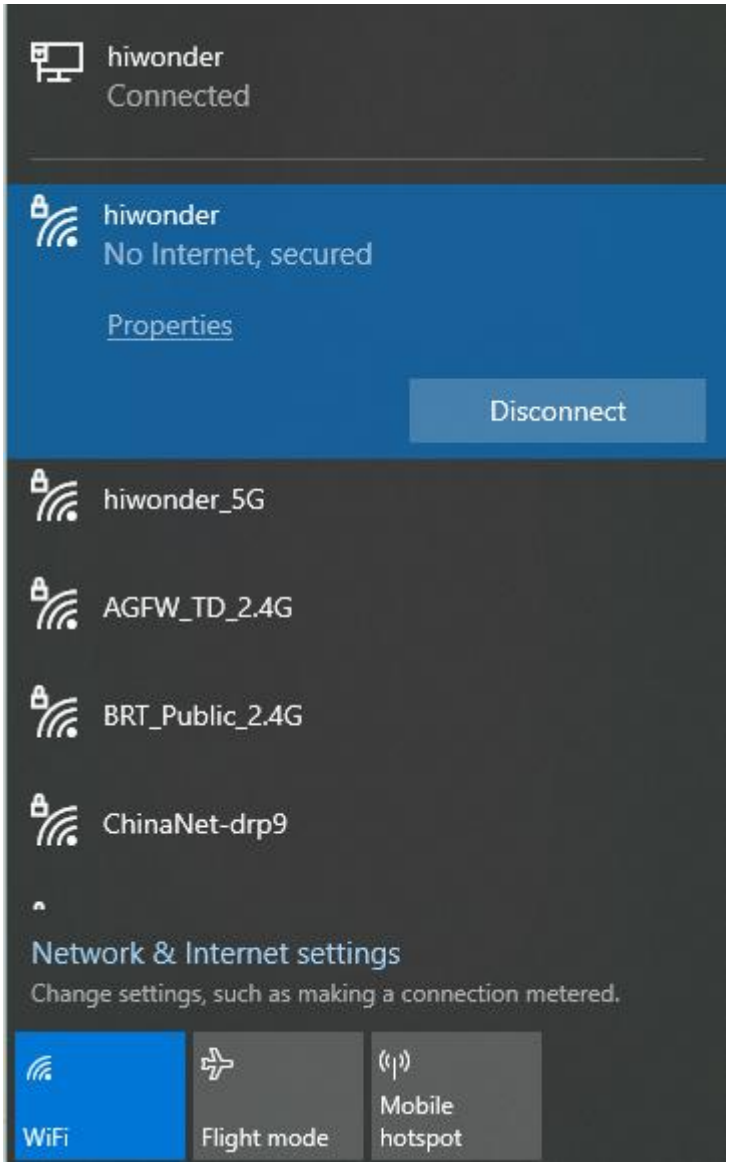


6) After installation, click the VNC icon

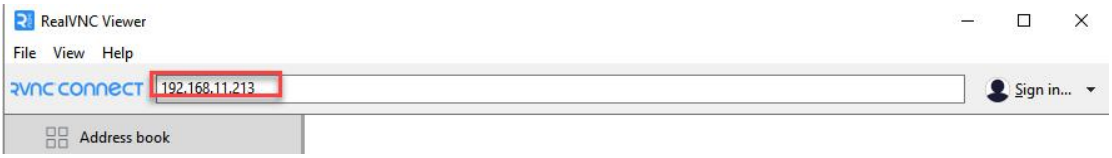
to open the software and start the connection.

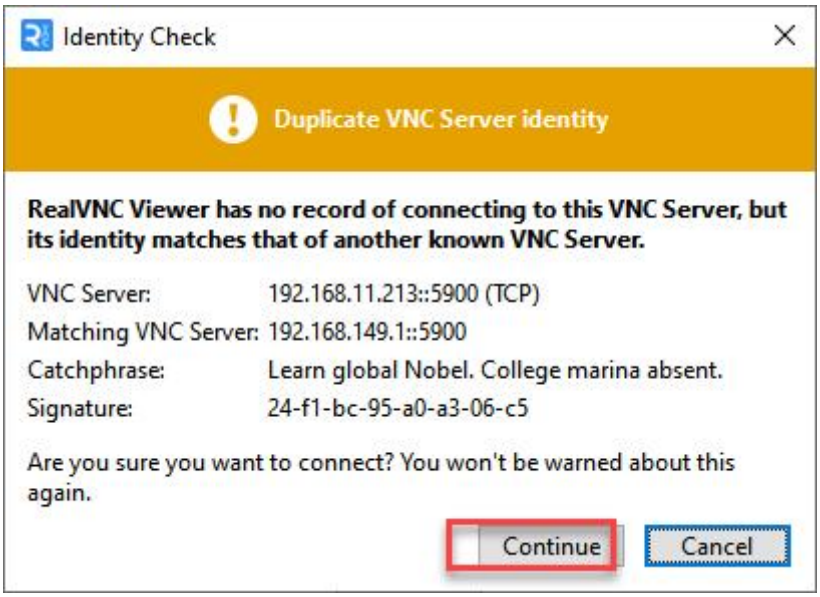
5.2 VNC Connection

1) Connect the Raspberry Pi 5 to the local network.For this tutorial, the Raspberry Pi 5 is already connected to the “Hiwonder” hotspot. Now, connect your computer to the same “Hiwonder” hotspot.



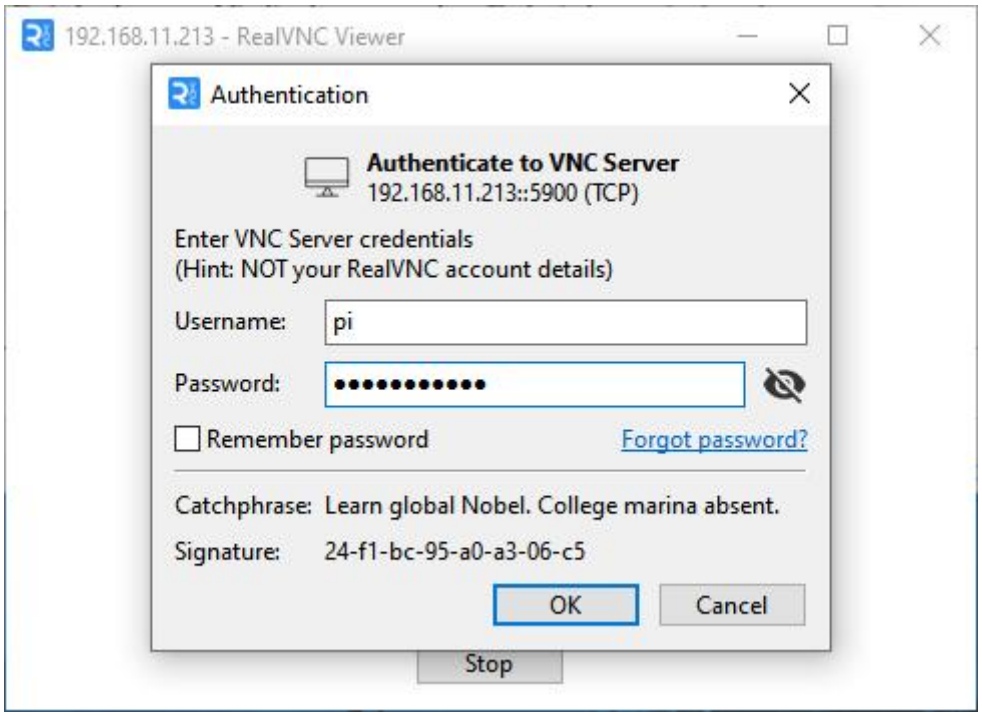
2) Launch the installed VNC client. In the VNC Viewer that opens, enter the Raspberry Pi 5's IP address: **192.168.11.199**, then press Enter. If a warning about an insecure connection appears, click "**Continue**."

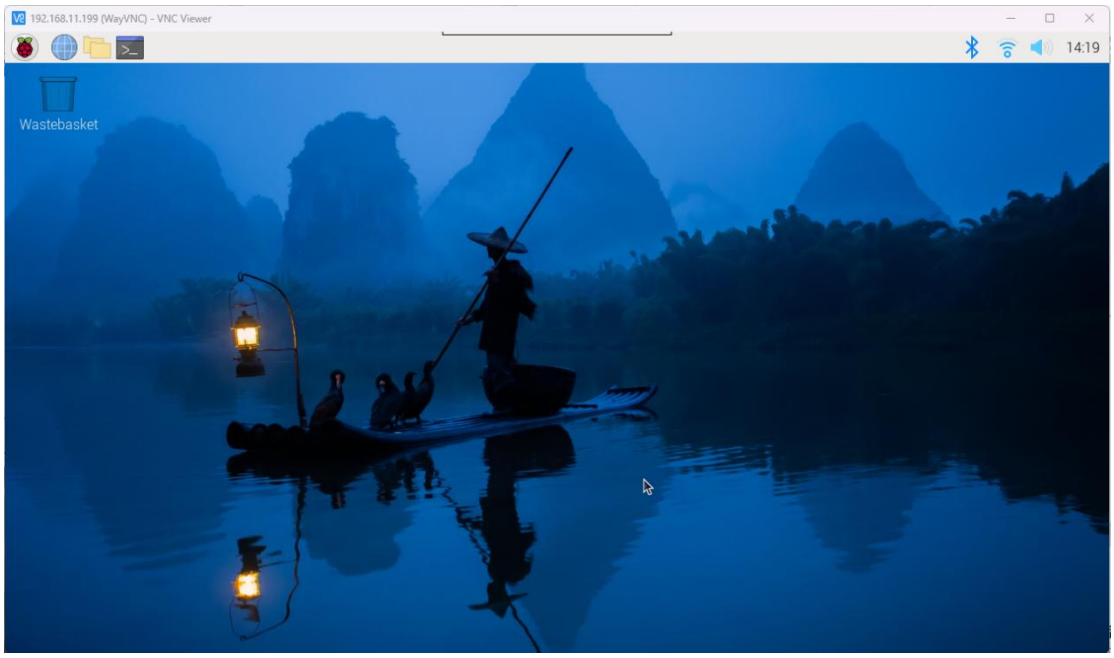




3) A prompt will appear asking for the password. Enter the password: "**raspberrypi**" (if prompted for a username, enter "pi"). Check the box to save the password, then click "OK." You will now be able to view the Raspberry Pi desktop remotely.

(If the screen is black with only the mouse pointer visible, try restarting the Raspberry Pi and repeating the steps.)




192.168.11.199 (WayVNC) - VNC Viewer
14:19
Wastebasket